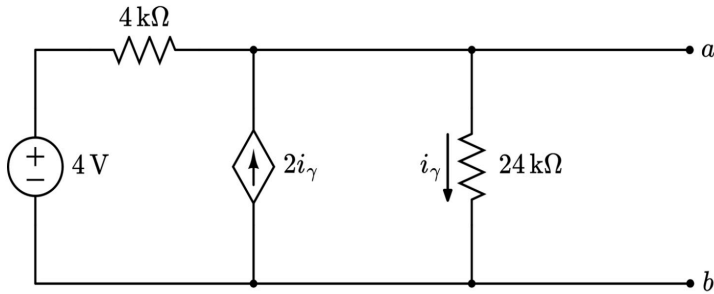


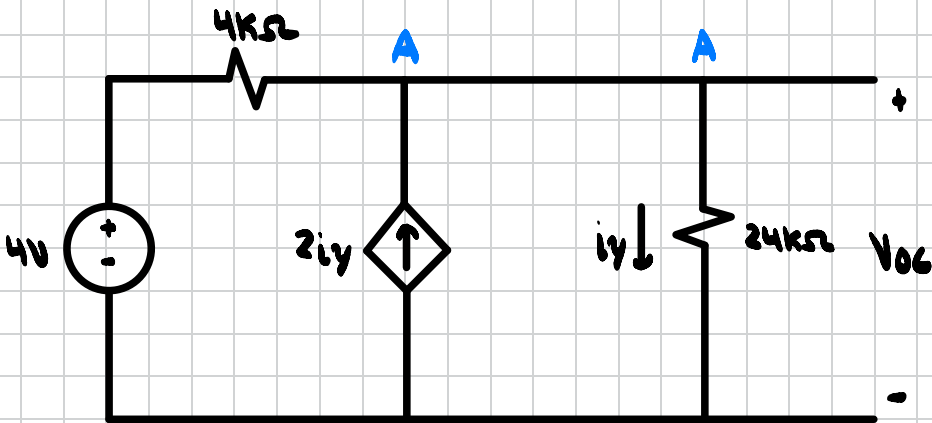
For the circuit shown below:

- Find the Thévenin equivalent circuit with respect to the open terminal pair ab and clearly illustrate the equivalent circuit with every value labeled.
- If a load is connected to points a and b , what is the maximum power that can be delivered to a load resistor R_L ?

Suggested time limit: 10 minutes



Voc Method



Step #1 Constraint

$$i_\gamma = \frac{V_{oc}}{24000}$$

Step #2 Nodal @ A

$$\frac{V_{oc} - 4}{4000} \cdot 2 + \frac{V_{oc} - 0}{24000} = 0$$

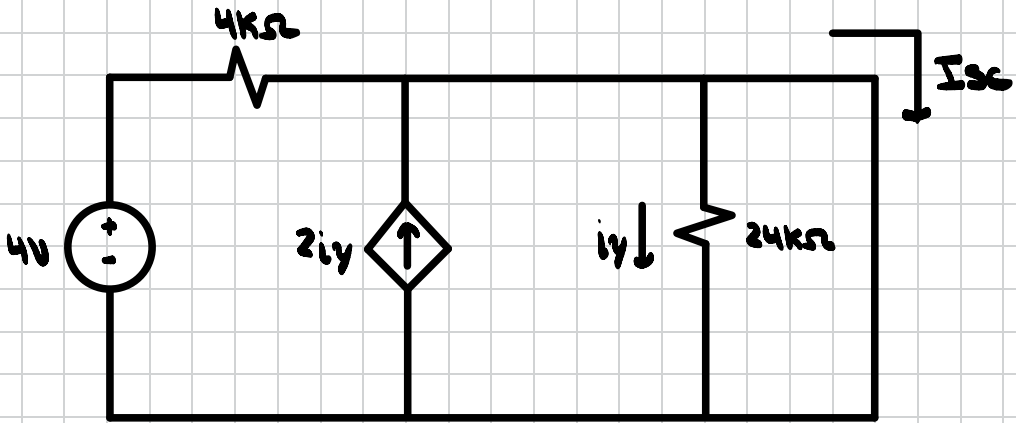
$$\frac{V_{oc} - 4}{4000} \cdot 2 \left(\frac{V_{oc}}{24000} \right) + \frac{V_{oc}}{24000} = 0$$

$$\frac{V_{oc}}{4000} - \frac{V_{oc}}{12000} + \frac{V_{oc}}{24000} = \frac{1}{10000}$$

$$\frac{1}{4800} V_{oc} = \frac{1}{10000}$$

$$V_{oc} = 4.8V = V_{TH}$$

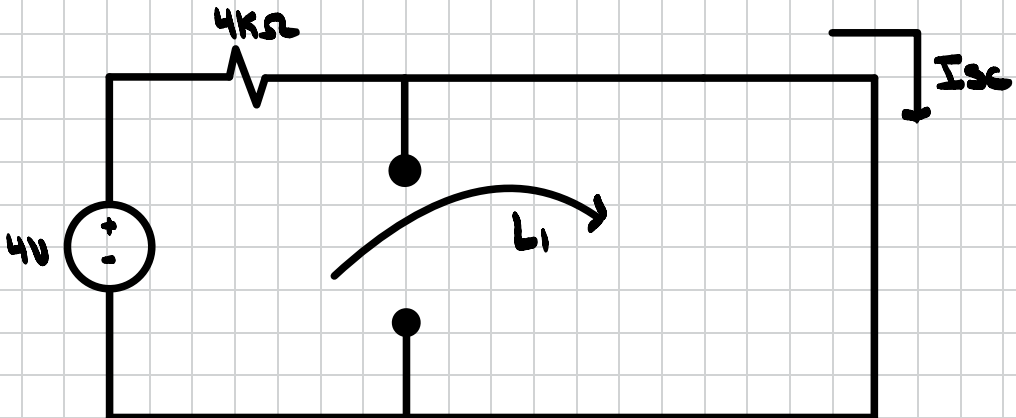
Isc Method



Step #1 Constraint

24000Ω Shorts so $i_y = 0A$ so dependent Source is $0A$
So Source can be replaced with an open circuit

Step #2 Redraw Circuit



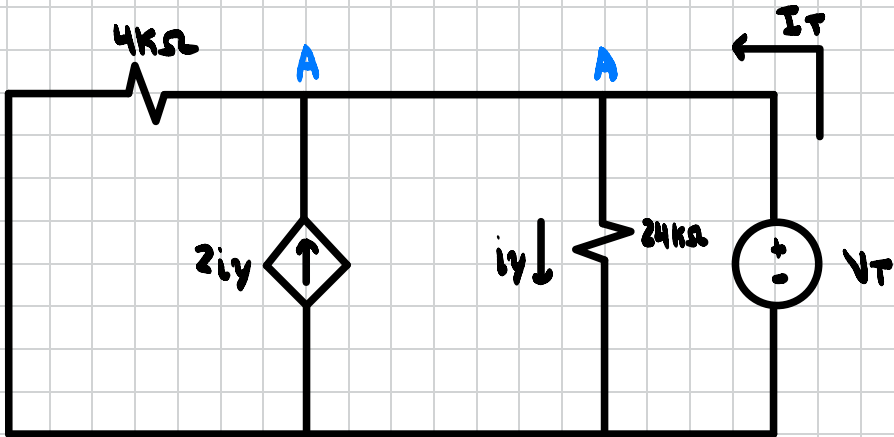
Step #3 KVL @ L_1

$$-4 + 4000I_{sc} = 0$$

$$4000I_{sc} = 4$$

$$I_{sc} = 1 \text{ mA} = I_N$$

Test Voltage Method



Step #1 Constraint

$$i_y = \frac{V_T}{24000}$$

Turn off all
independent Source
Voltage Source Short
Current Source open

Step #2 Nodal @ A

$$\frac{V_T}{4000} - 2i_y + \frac{V_T}{24000} - I_T = 0$$

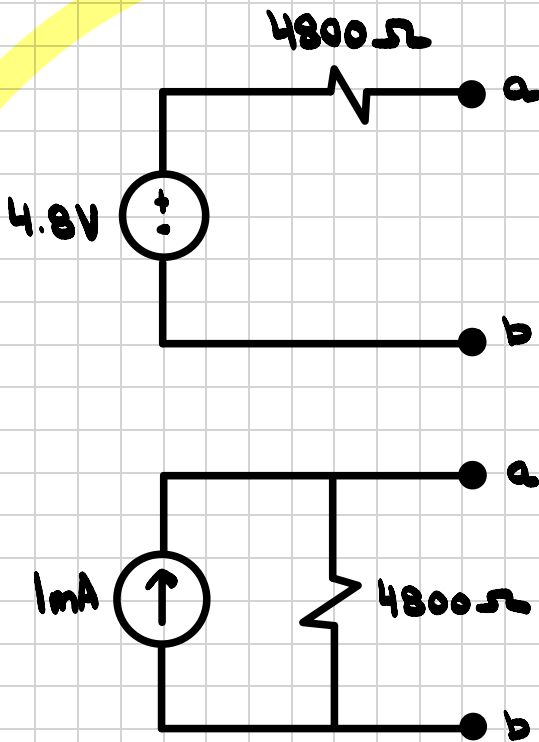
$$\frac{V_T}{4000} - 2\left(\frac{V_T}{24000}\right) + \frac{V_T}{24000} - I_T = 0$$

$$\frac{V_T}{4000} - \frac{V_T}{12000} + \frac{V_T}{24000} = I_T$$

$$\frac{V_T}{4800} = I_T$$

$$V_T = 4800 I_T$$

$$\frac{V_T}{I_T} = 4800\ \Omega = R_{TH}$$



Max Power Transfer

Step #1 Formula

$$P_{MAX} = \frac{(V_{TH})^2}{4R_{TH}}$$

$$R_L = R_{TH}$$

Step #2 Solve

$$P_{MAX} = \frac{(4.8)^2}{4(4800)}$$

$$P_{MAX} = 1.2 \text{ mW}$$